

Saman Nezafat

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Professional Summary

Backend Software Engineer with 4+ years of experience architecting high-concurrency systems and real-time platforms serving millions of users. Proven track record in technical leadership and delivering scalable, distributed solutions in Go and Python. Open to relocation with visa sponsorship.

Work Experience

Intelligent Processing of Pars Hadish - 2023-Present

- Stack: Go, Python, FastAPI, PostgreSQL, MongoDB, ClickHouse, Redis, RabbitMQ, WebSocket, Docker

Payment Gateway

- Architected a high-concurrency payment gateway in **Go**, ensuring transaction safety via **Redis distributed locking** and **idempotency**. Implemented a modular **adapter architecture** for seamless multi-provider integration.

Audit Log Service

- Built a tamper-evident audit log service in **Go** with **HMAC-SHA256 hash chains** per user, a **Redis Streams** → **ClickHouse** high-throughput ingestion pipeline, and an emergency mode with disk monitoring and automatic log rotation.

Meter Data Management System

- Developed a **Go** backend that collects meter data and runs a multi-tier billing engine to automatically generate detailed PDF invoices.

B2C Tourism Platform

- Designed and built a large-scale **tourism platform (3M+ users, 9+ B2B partners)** with **multi-gateway payment** orchestration and asynchronous, **cached** processing to reduce latency and database load.

Real-Time User Analytics System

- Built a high-throughput analytics system with a visual **Journey Builder** to automate personalized user flows, engagement triggers, and A/B tests; powered by **FastAPI**, **ClickHouse**, and **RabbitMQ**.

Messaging System

- Built a priority-managed messaging system (SMS & Push) using **Redis Streams** for efficient, real-time message queuing.

Real-Time Taxi POS Backend

- Developed a **WebSocket-powered** system for real-time **POS payments** and **live location streaming**.

AI-Powered Supermarket Agent

- Built an automated shopping assistant for WhatsApp, enabling users to place orders via voice or text, with the AI handling product search, cart management, and order creation.

Atishahr - 2022-2023

- Stack: Python, FastAPI, PostgreSQL

SIB Ticket System

- Optimized a key reporting endpoint, achieving a **20x performance** improvement and eliminating slowdowns under high load.
- Contributed to the project's second version using **FastAPI**, implementing **asynchronous features** to handle increased system load.

Mehr-e Pars ICT - 2021-2022

- Stack: Python, FastAPI, PostgreSQL

Organizational Resource Management System

- Engineered a **data migration** solution, enabling seamless transitions from legacy project management systems to the Rainesh platform.

Comprehensive Warehouse System

- Developed a **load-simulating reverse proxy** module for a large-scale warehouse system, simulating backend conditions under high load to improve frontend stability.

Open Source

- **Muxr**, A Go WebSocket multiplexing library for efficient real-time communications.
- **IP**, A Go TCP server that returns the client's IP and country in multiple formats.
- **PriceBot**, A Go Telegram bot providing real-time currency exchange rates, gold prices, and cryptocurrency values.
- **Social Token Experiment**, A minimalist ERC-20 social token on Polygon, designed as an experiment in on-chain communication (Solidity).
- **PyBotNet**, An educational security-research framework in Python for exploring remote-control patterns via Telegram.

Technical Skills

- **Languages**: Go, Python, Solidity
- **Frameworks & Protocols**: FastAPI, Gin, REST, WebSocket
- **Databases & Caching**: PostgreSQL, MongoDB, ClickHouse, Redis
- **Messaging & Streaming**: RabbitMQ, Redis Streams
- **Architecture**: Distributed systems, microservices, event-driven design, high concurrency, idempotency
- **DevOps**: Linux, Docker, Git, CI/CD
- **Security**: JWT, OWASP best practices

Soft Skills

- **Leadership & Collaboration**: Led a 3-member engineering team, guiding system design and code reviews while cultivating collaboration through mentoring and documentation.

Education

- **Bachelor of Science in Civil Engineering**

Languages

- **English**: Professional Working Proficiency